

Alexander X Yu

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Education

University of California, Los Angeles

B.S. in Computer Science
GPA: 4.0

Los Angeles, CA
September 2019 - June 2023

The Harker School

High School Diploma
Unweighted GPA: 4.0, Weighted GPA: 4.54

San Jose, CA
August 2015 - May 2019

Relevant coursework: AP Computer Science A, Data Structures, Expert Systems, Computer Architecture, Multivariable Calculus, Differential Equations, AP Statistics, AP Physics C: Mechanics and Electricity & Magnetism

Experience

Zili Lab

Research Assistant

UCLA Psychology Department
Oct 2019 - Present

- Used support vector machines and neural networks to study directional motion processing in the human brain
- Coded interfaces for stimulus-based human behavioral experiments
- Evaluated and simulated human decision-rule models based on same-different experiments

Li Lab

Research Assistant

UCSF Department of Biochemistry and Biophysics
June 2018 - Sep 2019

- Used machine learning algorithms and image/data analysis libraries (Tensorflow, Scikit, SciPy, Pandas, Matplotlib, etc.) to investigate causal links between the cell cycle regulator WHI5 and aging process in yeast
- Built computational pipelines to analyze physical differences (body mass, metabolic rate) and genomic differences (Gene Ontology, positive selection in genes, etc.) between mammalian species that may lead to variation in lifespan

Skills

Programming Languages: Java, Python, C++, Bash, HTML, CSS, Javascript, Jess/LISP, R, SQL, and C

Software Tools: Jupyter Notebook, Postman, MongoDB

Extracurricular Positions: UCLA Symphony Cellist (2019-present), Harker Swim Team Captain (2019), Harker Orchestra Assistant Principal Cellist (2018-19)

Projects

Footprint

An intuitive app that allows people to track and visualize their carbon footprint.

Javascript, Node.js, React

Music Scroll

A sheet music viewer that automatically switches pages hands-free.

Javascript, Node.js

SMEP

An expert system that helps to choose events to enter for a given swim meet.

JESS, Java

Awards

- 2019 **National Merit Scholar**, Recognized for academic and extracurricular accomplishments.
- 2019 **Presidential Scholar Nominee**, Recognized as a distinguished high school senior.
- 2017 **USA Computing Olympiad Silver Level**, Awarded to those who have reached the USACO silver level.